

X is a R.V. with probability density function (PDF)

$$\text{Ex: } f_X(x) = \begin{cases} \frac{10}{x^2}, & x > 10 \\ 0, & x \leq 10 \end{cases}$$

Find cumulative distribution function (CDF) of x .
 $P(15 < X < 20)$

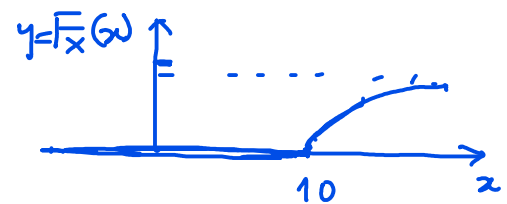
Soln: $F_X(x) = P(X \leq x)$

$$= \int_{-\infty}^x f_X(t) dt$$

If $x \leq 10$, $F_X(x) = 0$

$$\begin{aligned} \text{If } x > 10, \quad F_X(x) &= \int_{10}^x \frac{10}{t^2} dt \\ &= \left[-\frac{10}{t} \right]_{10}^x \\ &= 1 - \frac{10}{x} \end{aligned}$$

$$\therefore F_X(x) = \begin{cases} 0, & x \leq 10 \\ 1 - \frac{10}{x}, & x > 10 \end{cases}$$



$$P(15 < X < 20) = \int_{15}^{20} f_X(t) dt$$

$$= \int_{15}^{20} \frac{10}{t^2} dt$$

$$= -\frac{10}{t} \Big|_{15}^{20}$$

$$= 10 \left(\frac{1}{15} - \frac{1}{20} \right) = \frac{1}{6}$$

$$P(15 < X < 20) = P(X < 20) - P(X \leq 15)$$

$$= P(X \leq 20) - P(X \leq 15)$$

$$= F_X(20) - F_X(15) = 1 - \frac{10}{20} - \left(1 - \frac{10}{15} \right)$$

$$= 10\left(\frac{1}{15} - \frac{1}{20}\right) = \frac{1}{6} \checkmark$$

X R.V. $P_X(x_i) = P(X=x_i)$

$$E(X) = \begin{cases} \sum_i x_i P_X(x_i), & \text{if } X \text{ is discrete RV} \\ \int_{-\infty}^{\infty} x f_X(x) dx, & \text{if } X \text{ is a continuous RV} \end{cases}$$

Ex.

$$P_X(0) = \frac{1}{8}, P_X(1) = \frac{3}{8}, P_X(2) = \frac{3}{8}, P_X(3) = \frac{1}{8}$$

$$\underline{\mu = E(X)} = \sum_i x_i P_X(x_i)$$

$$= 0 \cdot \frac{1}{8} + 1 \cdot \frac{3}{8} + 2 \cdot \frac{3}{8} + 3 \cdot \frac{1}{8} = \boxed{1.5} \checkmark$$

Ex:

$$E(X) = \int_{-\infty}^{\infty} x f_X(x) dx$$

$$\underline{\int_{-\infty}^{\infty} |x| f_X(x) dx \text{ is convergent}}$$

$$= \int_0^{\infty} x e^{-x} dx$$

$$= [x(-e^{-x}) - (1)(e^{-x})]_0^{\infty}$$

$$= 1$$

Ex: X is a R.V with PDF $f_X(x) = \frac{1}{\pi} \frac{1}{1+x^2}, -\infty < x < \infty$

Check whether $E(X)$ exists.

$$\int_{-\infty}^{\infty} f_X(x) dx = \frac{1}{\pi} \int_{-\infty}^{\infty} \frac{1}{1+x^2} dx = \frac{1}{\pi} [\tan^{-1} x]_{-\infty}^{\infty} = 1$$

$$\frac{1}{\pi} \int_{-\infty}^{\infty} |x| \frac{1}{1+x^2} dx$$

$$\int_0^{\infty} \frac{x}{1+x^2} dx = \frac{1}{2} \int_0^{\infty} \frac{2x}{1+x^2} dx = \frac{1}{2} \lim_{b \rightarrow \infty} \int_0^b \frac{2x}{1+x^2} dx$$

$$= \frac{1}{2} \lim_{b \rightarrow \infty} [\log(1+x^2)]_0^b$$

$\therefore E(X)$ does not exist.